

Supervised Learning

K-Nearest Neighbors

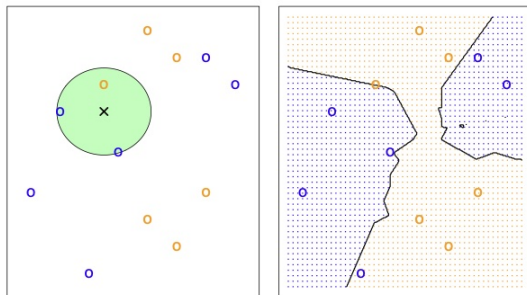
KNN classifier

- The response variable Y is categorical.
- Choose a positive integer K .
- Let $\mathbf{x}^0 \in \mathbb{R}^p$ be a test observation for which we want to predict the qualitative response Y . Identify $\mathcal{N}_0$ as the set consisting of the K points in the training data that are closest to $\mathbf{x}^0$.
- Estimate the conditional probability for class j , by

$$Pr(Y = j | \mathbf{X} = \mathbf{x}^0) = \frac{1}{K} \sum_{i \in \mathcal{N}_0} I(y_i = j).$$

- The KNN classifies the test observation $\mathbf{x}^0$ to the class with the largest probability value.

KNN classifier



- Two classes: **blue** and **orange**;
- Number of neighbors $K = 3$;
- $Pr(Y = 1|\mathbf{X} = \mathbf{x}^0) = \frac{1}{3} \sum_{i \in \mathcal{N}_0} I(y_i = 1)$.

KNN regression

- The KNN classifier and regressor fundamentally rely on a distance metric. In general, the Euclidean distance is used,

$$d(\mathbf{x}^0, \mathbf{x}^i) = \|\mathbf{x}^0 - \mathbf{x}^i\|_2 = \sqrt{\sum_{j=1}^p (x_j^0 - x_j^i)^2}.$$

- In regression, we want to predict the target function $f(\mathbf{x})$, $f : \mathbb{R}^p \rightarrow \mathbb{R}$, which shapes the relationship between the response y and the predictors $\mathbf{x} \in \mathbb{R}^p$, $y = f(\mathbf{x}) + \epsilon$. The KNN regressor will compute the prediction $\hat{y}_0 = \hat{f}(\mathbf{x}^0) \approx y_0$ using the formula

$$\hat{y}_0 = \frac{1}{K} \sum_{i \in \mathcal{N}_0} y_i$$

K-Nearest Neighbors

- Scaling the data is an important preprocessing step for KNN because the algorithm is distance-based;
- KNN is computationally intensive;
- **(Curse of Dimensionality)** Assume we keep the number of data points (n) fixed but we increase the number of predictors (p). For a test point $\mathbf{x}^0$, the volume of the neighborhood $\mathcal{N}_0$ capturing a fixed number K of neighbors, grows larger and larger as p increases $\Rightarrow$ the “neighbors” become less and less “similar” to the query point (as p increases).

K-Nearest Neighbors

Example (curse of dimensionality).

Consider all data points $\mathbf{x} \in [0, 1]^p$. Let $K = 3$, $n = 100$ and all training data $\mathbf{x}^i$ uniformly distributed in $[0, 1]^p$. Let $\mathbf{x}^0 \in [0, 1]^p$ be a query point.

- $p = 1$. To get the 3 nearest neighbors of $\mathbf{x}^0$, we expect to cover $3/100 = 0.03$ of $[0, 1]$ ($0.03 \times 100\% = 3\%$).

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- $p = 10$. To get the 3 nearest neighbors of $\mathbf{x}^0$, we expect to cover $0.03^{1/10}$ of $[0, 1]^{10}$ ($0.03^{1/10} \times 100\% \approx 70.4\%$).